



ROBOCAR

MASK GENERATOR



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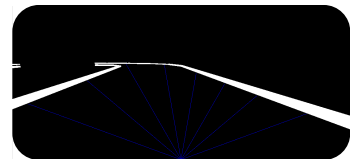
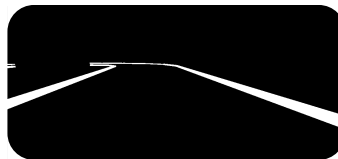
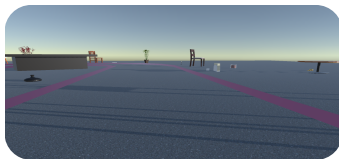
Language: Python, C#

What is Mask Generator?

Now that you created the AI that controls the car, you will need to create the AI that generate the lines from the image.

As you saw in the RacingSimulator, we only keep the lines from the image of the camera before creating the 2D raycast.

The goal of this project is to discover the unity part of the simulation. This will allow you to create an image/mask generator in order to create a line recognition AI and then raycast projection. Here are the step (from an older simulation):



- ✓ Try to do most of the calculations on your GPU to speed up the full process from image to mask of the line with raycast
- ✓ If your pc lacks power, look on Kaggle

Project Structure

This project is divided into three main components:

- ✓ A **image/mask generator**
- ✓ A **line recognition AI**
- ✓ A **2D raycast projection**

Unity



- ✓ The unity project is not perfect, both code and game objects can be upgraded but it is not the priority for now

Download zip [here](#)

Install Unity Hub.

Unzip the project, open Unity Hub and add a project (the one you unzipped), it will ask you to install the corresponding editor version, do it.

You will maybe have some errors, but it should be missing lib so go to the window manager and install them from the unity registry.

Now you have access to the simulation, take your time to understand how it work, you can modify it as you wish but be careful not to waste too much time on unnecessary things until you finished the project.

Generator

First thing to do is to create a dataset to make your AI learn to generate a mask from an image. This dataset should associate images of the camera with mask (images containing the lines only).



- ✓ Don't forget about data augmentation, there are a lot to do
- ✓ Try to have at least 5000 pairs (image + corresponding mask)

Line Recognition AI

Create an AI that generates masks from images like in the dataset.

You should look for Convolutional Neural Networks (CNN) models, some are specialized in object extraction.



- ✓ Quantize your model to make it usable on a 32 bit architecture so that you can use it on the car later
- ✓ It is important to have a quick inference

2D Raycast

In order to simplify the data we send to the AI which controls the car we use a 2D raycast. It is way lighter than a full picture and it is easier for the model to understand the relation between the inputs and outputs.

You need to create an algorithm which creates those rays and again you need it to be as fast as possible.



- ✓ Try having parameters for field of view and number of rays for more scalability

Bonus Features

If everything works as expected, consider implementing the following improvements, it will help you for the following steps of the project:

- ✓ Reinforcement learning for the AI that controls the car
- ✓ You can try modifying the unity project to send different data to your AI (full image instead of raycast, 360° raycast, ...) but we will wait a solid explanation while reviewing
- ✓ Upgrade the simulation

{EPITECH}

